

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14 COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

Assessment Tool Weightage: 40%

1.A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.

Commands and outcome:

1. ping -c 4 scanme.nmap.org:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali㉿kali)-[~]
$ ping -c 4 scanme.nmpa.org
PING scanme.nmpa.org (141.193.213.10) 56(84) bytes of data.
64 bytes from 141.193.213.10: icmp_seq=1 ttl=64 time=28.4 ms
64 bytes from 141.193.213.10: icmp_seq=2 ttl=64 time=12.9 ms
64 bytes from 141.193.213.10: icmp_seq=3 ttl=64 time=13.5 ms
64 bytes from 141.193.213.10: icmp_seq=4 ttl=64 time=38.7 ms

— scanme.nmpa.org ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 12.945/23.372/38.666/10.796 ms
```

2. `sudo nmap -sn scanme.nmap.org`

```
(kali㉿kali)-[~]
└─$ sudo nmap -sn scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 00:57 -0400
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Failed to resolve "scanme.nmap.org".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 4.73 seconds
```

3. `sudo nmap -sT scanme.nmap.org`

```
(kali㉿kali)-[~]
└─$ sudo nmap -sT scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 00:58 -0400
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Failed to resolve "scanme.nmap.org".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 4.69 seconds
```

4. `sudo nmap -sV scanme.nmap.org`

```
(kali㉿kali)-[~]
└─$ sudo nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 00:59 -0400
Failed to resolve "scanme.nmap.org".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.07 seconds
```

5. `sudo nmap -O scanme.nmap.org`

```
(kali㉿kali)-[~]
└─$ sudo -O scanme.nmap.org
sudo: invalid option -- 'O'
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
        [-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
        [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
        [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
        [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
        [-u user] file ...
```

6. `sudo nmap -sC scanme.nmap.org`

```
(kali㉿kali)-[~]
$ sudo nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 01:00 -0400
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:06 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:09 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 35.27% done; ETC: 01:00 (0:00:17 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.035s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 986 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
500/tcp   closed isakmp
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
8089/tcp  closed unknown

Nmap done: 1 IP address (1 host up) scanned in 57.38 seconds
```

7. `sudo nmap -sC -sV -O scanme.nmap.org`

```
(kali㉿kali)-[~]
$ sudo nmap -sC -sV -O scanme.namp.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 01:04 -0400
Failed to resolve "scanme.namp.org".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 5.07 seconds
```

8. `sudo nmap -p- -sV -O scanme.nmap.org`

```
(kali㉿kali)-[~]
$ sudo nmap -p- -sV -O scanme.namp.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 01:07 -0400
Failed to resolve "scanme.namp.org".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 7.60 seconds
```

2.The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies

Commands and outcome:

1. ping -c 4 192.168.56.101

```
(kali㉿kali)-[~]
└─$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
ping: Warning: invalid tv_usec -88005735319754866 us
ping: Warning: time of day goes back (-203731396528950 s), taking countermeasures
ping: Warning: invalid tv_usec -88005735319748209 us
ping: Warning: time of day goes back (-203731396528950 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.000 ms
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 65 8 0 aa fb 16 60 0 3 f f5 7a 6a 0 0 0 0 e6 6e 7 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319747749 us
ping: Warning: time of day goes back (-203731396528950 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 65 8 0 aa fb 16 60 0 3 f f5 7a 6a 0 0 0 0 e6 6e 7 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319747595 us
ping: Warning: time of day goes back (-203731396528950 s), taking countermeasures
64 bytes from 192.168.56.101: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 65 8 0 aa fb 16 60 0 3 f f5 7a 6a 0 0 0 0 e6 6e 7 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b

— 192.168.56.101 ping statistics —
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 3160ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4
```

2. ping -sn 192.168.56.101

```
(kali㉿kali)-[~]
└─$ sudo nmap -sn 192.168.56.101
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:14 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0012s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.08 seconds
```

3. `sudo nmap -sT 192.168.56.101`

```
(kali㉿kali)-[~]
$ sudo nmap -sT 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:15 -0400
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 51.30% done; ETC: 06:15 (0:00:09 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0073s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 15.66 seconds
```

4. `sudo nmap -p- 192.168.56.101`

```
(kali㉿kali)-[~]
$ sudo nmap -p- 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:22 -0400
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.50% done; ETC: 06:27 (0:04:23 remaining)
Stats: 0:01:19 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 23.11% done; ETC: 06:28 (0:04:16 remaining)
Stats: 0:05:46 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 60.91% done; ETC: 06:32 (0:03:41 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.032s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 594.93 seconds
```

5. `sudo nmap -sV 192.168.56.101`

```
(kali㉿kali)-[~]
$ sudo nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:33 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0036s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 75.35 seconds
```


6. sudo nmap -O 192.168.56.101

```
(kali@kali)-[~]
$ sudo nmap -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:37 -0400
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.56.101
Host is up (0.0047s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.14 seconds
```

7. sudo nmap -sC 192.168.56.101

```
(kali@kali)-[~]
$ sudo nmap -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:41 -0400
Stats: 0:00:49 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.59% done; ETC: 06:42 (0:00:00 remaining)
Stats: 0:00:54 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.59% done; ETC: 06:42 (0:00:00 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0031s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 71.38 seconds
```

8. sudo nmap -p- -sV -O -sC 192.168.56.101

```
(kali@kali)-[~]
$ sudo nmap -p- -sV -O -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 06:54 -0400
Stats: 0:01:56 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 35.80% done; ETC: 06:59 (0:03:24 remaining)
Stats: 0:02:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 66.96% done; ETC: 06:58 (0:01:27 remaining)
Stats: 0:03:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.98% done; ETC: 06:58 (0:01:10 remaining)
Stats: 0:04:20 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 68.75% done; ETC: 06:58 (0:00:01 remaining)
Stats: 0:05:03 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 93.75% done; ETC: 06:59 (0:00:03 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.088s latency).
Not shown: 58632 filtered tcp ports (no-response), 6901 filtered tcp ports (net-unreach)
PORT      STATE SERVICE  VERSION
5060/tcp  open  tcpwrapped
5061/tcp  open  tcpwrapped
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridge|VoIP adapter|general purpose|media device
Running (JUST GUESSING): Oracle Virtualbox (97%), Slirp (97%), AT&T embedded (94%), QEMU (94%), Sanyo embedded (87%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu cpe:/h:sanyo:plc-xu88
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (97%), AT&T BGW210 voice gateway (94%), QEMU user mode network gateway (87%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 320.76 seconds
```

3.A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

Commands and outcome:

1. ping -c 4 192.168.56.102

```
(kali㉿kali)-[~]
$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
ping: Warning: invalid tv_usec -88005735319661157 us
ping: Warning: time of day goes back (-151066507542125 s), taking countermeasures
ping: Warning: invalid tv_usec -88005735319657372 us
ping: Warning: time of day goes back (-151066507542125 s), taking countermeasures
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 66 8 0 8b 5d 16 61 0 3 d8 1 7b 6a 0 0 0 0 39 ff a 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319657301 us
ping: Warning: time of day goes back (-151066507542125 s), taking countermeasures
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 66 8 0 8b 5d 16 61 0 3 d8 1 7b 6a 0 0 0 0 39 ff a 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319657059 us
ping: Warning: time of day goes back (-151066507542125 s), taking countermeasures
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 66 8 0 8b 5d 16 61 0 3 d8 1 7b 6a 0 0 0 0 39 ff a 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b

— 192.168.56.102 ping statistics —
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 3235ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4
```

2. sudo nmap -sn 192.168.56.102

```
(kali㉿kali)-[~]
$ sudo nmap -sn 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:05 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00055s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.35 seconds
```

3. sudo nmap -p- 192.168.56.102

```
(kali㉿kali)-[~]
└─$ sudo nmap -p- 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:06 -0400
Stats: 0:00:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.80% done; ETC: 07:10 (0:04:32 remaining)
Stats: 0:03:07 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.16% done; ETC: 07:09 (0:00:32 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.072s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 221.48 seconds
```

4. sudo nmap -sV 192.168.56.102

```
(kali㉿kali)-[~]
└─$ sudo nmap -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:10 -0400
Stats: 0:00:19 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Stats: 0:00:42 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Nmap scan report for 192.168.56.102
Host is up (0.0028s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 76.93 seconds
```

5. sudo nmap -O 192.168.56.102

```
(kali㉿kali)-[~]
└─$ sudo nmap -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:13 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0045s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.09 seconds
```

6. sudo nmap -sC 192.168.56.102

```
(kali㉿kali)-[~]
└─$ sudo nmap -sC 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:15 -0400
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.18% done; ETC: 07:16 (0:00:00 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.0031s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 69.57 seconds
```


7. `sudo nmap -p 139,445 --script smb-protocols 192.168.56.102`

```
(kali@kali)-[~]
$ sudo nmap -p 139,445 --script smb-protocols 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:19 -0400
NSE: failed to initialize the script engine:
/usr/share/nmap/nse_main.lua:829: 'smb-protocols' did not match a category, filename, or directory
stack traceback:
  [C]: in function 'error'
  /usr/share/nmap/nse_main.lua:829: in local 'get_chosen_scripts'
  /usr/share/nmap/nse_main.lua:1364: in main chunk
  [C]: in ?

QUITTING!
```

8. `sudo nmap -p 445 --script smb-security-mode 192.168.56.102`

```
(kali@kali)-[~]
$ sudo nmap -p 445 --script smb-security-mode 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:22 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00100s latency).

PORT      STATE      SERVICE
445/tcp    filtered  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 3.98 seconds
```

9. `sudo nmap -p- -sV -O -sC 192.168.56.102`

```
(kali@kali)-[~]
$ sudo nmap -p- -sV -O -sC 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 07:23 -0400
Stats: 0:02:57 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 56.21% done; ETC: 07:28 (0:02:15 remaining)
Stats: 0:03:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 58.13% done; ETC: 07:29 (0:02:15 remaining)
Stats: 0:05:48 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 0.00% done
Stats: 0:05:54 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.96% done; ETC: 07:29 (0:00:00 remaining)
Stats: 0:05:54 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.31% done; ETC: 07:29 (0:00:00 remaining)
Stats: 0:06:01 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.31% done; ETC: 07:29 (0:00:00 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.011s latency).
Not shown: 49021 filtered tcp ports (no-response), 16512 filtered tcp ports (net-unreach)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 425.34 seconds
```

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure.

Commands and outcome:

1. `ping -c 4 192.168.56.103`

```
(kali㉿kali)-[~]
$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.
ping: Warning: invalid tv_usec -88005735319779557 us
ping: Warning: time of day goes back (-264634032630910 s), taking countermeasures
ping: Warning: invalid tv_usec -88005735319772196 us
ping: Warning: time of day goes back (-264634032630910 s), taking countermeasures
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 67 8 0 4 4d 2e 75 0 3 c7 53 7d 6a 0 0 0 0 be a9 3 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319772151 us
ping: Warning: time of day goes back (-264634032630910 s), taking countermeasures
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 67 8 0 4 4d 2e 75 0 3 c7 53 7d 6a 0 0 0 0 be a9 3 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319772126 us
ping: Warning: time of day goes back (-264634032630910 s), taking countermeasures
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 67 8 0 4 4d 2e 75 0 3 c7 53 7d 6a 0 0 0 0 be a9 3 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b

— 192.168.56.103 ping statistics —
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 3489ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4
```

2. `sudo nmap -sn 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -sn 192.168.56.103
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:19 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0016s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds
```

3. `sudo nmap -p- 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -p- 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:20 -0400
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.94% done; ETC: 01:24 (0:04:28 remaining)
Stats: 0:02:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.12% done; ETC: 01:25 (0:03:31 remaining)
Stats: 0:02:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.60% done; ETC: 01:25 (0:03:11 remaining)
Stats: 0:06:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 93.74% done; ETC: 01:27 (0:00:27 remaining)
Nmap scan report for 192.168.56.103
Host is up (0.084s latency).
All 65535 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 427.11 seconds
```

4. `sudo nmap -sV 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -sV 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:27 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0016s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.99 seconds
```

5. `sudo nmap -sC 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -sC 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:28 -0400
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 88.89% done; ETC: 01:28 (0:00:01 remaining)
Nmap scan report for 192.168.56.103
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 12.16 seconds
```

6. `sudo nmap -p 80 --script http-title,http-server-header 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -p 80 --script http-title,http-server-header 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:30 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00029s latency).

PORT      STATE      SERVICE
80/tcp    filtered  http

Nmap done: 1 IP address (1 host up) scanned in 1.22 seconds
```

7. `sudo nmap -p 3000 --script http-title,http-server-header 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -p 3000 --script http-title,http-server-header 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:31 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0045s latency).

PORT      STATE      SERVICE
3000/tcp   filtered   ppp

Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds
```

8. `sudo nmap -p 80,443,3000 -sV 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -p 80,443,3000 -sV 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 01:33 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0016s latency).

PORT      STATE      SERVICE VERSION
80/tcp    filtered   http
443/tcp    filtered   https
3000/tcp   filtered   ppp

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.50 seconds
```

9. `sudo nmap -p- -sV -O -sC 192.168.56.103`

```
(kali㉿kali)-[~]
$ sudo nmap -p- -sV -O -sC 192.168.56.103
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
```


version trace: Show detailed version scan activity (for debugging)

SCRIPT SCAN:

- sC: equivalent to --script=default
- script=<Lua scripts>: <Lua scripts> is a comma separated list of directories, script-files or script-categories
- script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
- script-args-file=filename: provide NSE script args in a file
- script-trace: Show all data sent and received
- script-updatedb: Update the script database.
- script-help=<Lua scripts>: Show help about scripts.
<Lua scripts> is a comma-separated list of script-files or script-categories.

OS DETECTION:

- O: Enable OS detection
- osscan-limit: Limit OS detection to promising targets
- osscan-guess: Guess OS more aggressively

TIMING AND PERFORMANCE:

Options which take <time> are in seconds, or append 'ms' (milliseconds), 's' (seconds), 'm' (minutes), or 'h' (hours) to the value (e.g. 30m).

- T<0-5>: Set timing template (higher is faster)
- min-hostgroup/max-hostgroup <size>: Parallel host scan group sizes
- min-parallelism/max-parallelism <numprobes>: Probe parallelization
- min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time>: Specifies probe round trip time.
- max-retries <tries>: Caps number of port scan probe retransmissions.
- host-timeout <time>: Give up on target after this long
- scan-delay/--max-scan-delay <time>: Adjust delay between probes
- min-rate <number>: Send packets no slower than <number> per second
- max-rate <number>: Send packets no faster than <number> per second

FIREWALL/IDS EVASION AND SPOOFING:

- f; --mtu <val>: fragment packets (optionally w/given MTU)
- D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
- S <IP_Address>: Spoof source address
- e <iface>: Use specified interface
- g/--source-port <portnum>: Use given port number
- proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies
- data <hex string>: Append a custom payload to sent packets
- data-string <string>: Append a custom ASCII string to sent packets
- data-length <num>: Append random data to sent packets
- ip-options <options>: Send packets with specified ip options
- ttl <val>: Set IP time-to-live field
- spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
- badsum: Send packets with a bogus TCP/UDP/SCTP checksum

OUTPUT:

- oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<rIpt kIddi3, and Grepable format, respectively, to the given filename.
- oA <basename>: Output in the three major formats at once
- v: Increase verbosity level (use -vv or more for greater effect)
- d: Increase debugging level (use -dd or more for greater effect)
- reason: Display the reason a port is in a particular state
- open: Only show open (or possibly open) ports
- packet-trace: Show all packets sent and received
- iflist: Print host interfaces and routes (for debugging)
- append-output: Append to rather than clobber specified output files
- resume <filename>: Resume an aborted scan
- noninteractive: Disable runtime interactions via keyboard
- stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
- webxml: Reference stylesheet from Nmap.Org for more portable XML
- no-stylesheet: Prevent associating of XSL stylesheet w/XML output

MISC:

- 6: Enable IPv6 scanning
- A: Enable OS detection, version detection, script scanning, and traceroute
- datadir <dirname>: Specify custom Nmap data file location
- send-eth/--send-ip: Send using raw ethernet frames or IP packets
- privileged: Assume that the user is fully privileged
- unprivileged: Assume the user lacks raw socket privileges
- V: Print version number
- h: Print this help summary page.

EXAMPLES:

```
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
```

SEE THE MAN PAGE (<https://nmap.org/book/man.html>) FOR MORE OPTIONS AND EXAMPLES

Scantype v not supported

5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise

Commands and outcome:

1. ping -c 4 192.168.56.1

```
(kali㉿kali)-[~]
$ ping -c 4 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.
64 bytes from 192.168.56.1: icmp_seq=1 ttl=64 time=9.23 ms
64 bytes from 192.168.56.1: icmp_seq=2 ttl=64 time=28.0 ms
64 bytes from 192.168.56.1: icmp_seq=3 ttl=64 time=27.6 ms
64 bytes from 192.168.56.1: icmp_seq=4 ttl=64 time=1.95 ms

— 192.168.56.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.948/16.711/28.034/11.418 ms
```

2. sudo nmap -sn 192.168.56.1

```
(kali㉿kali)-[~]
$ sudo nmap -sn 192.168.56.1
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:20 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0017s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.55 seconds
```

3. sudo nmap -p- 192.168.56.1

```
(kali@kali)-[~]
$ sudo nmap -p- 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:20 -0400
Stats: 0:00:39 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 25.01% done; ETC: 02:23 (0:01:51 remaining)
Stats: 0:01:39 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 87.76% done; ETC: 02:22 (0:00:13 remaining)
Nmap scan report for 192.168.56.1
Host is up (0.0046s latency).
Not shown: 65518 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
137/tcp    closed netbios-ns
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
5040/tcp   open  unknown
33060/tcp  open  mysqlx
49664/tcp  open  unknown
49665/tcp  open  unknown
49666/tcp  open  unknown
49667/tcp  open  unknown
49668/tcp  open  unknown
49672/tcp  open  unknown
49675/tcp  open  unknown
57621/tcp  open  unknown
62562/tcp  open  unknown
63176/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 109.10 seconds
```

4. sudo nmap -sV 192.168.56.1

```
(kali@kali)-[~]
$ sudo nmap -sV 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:23 -0400
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 17.80% done; ETC: 02:23 (0:00:14 remaining)
Nmap scan report for 192.168.56.1
Host is up (0.00048s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3306/tcp   open  tcpwrapped
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.04 seconds
```

5. sudo nmap -O 192.168.56.1

```
(kali@kali)-[~]
$ sudo nmap -O 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:24 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00086s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.24 seconds
```

6. sudo nmap -sC 192.168.56.1

```
(kali㉿kali)-[~]
$ sudo nmap -sC 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:24 -0400
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 93.92% done; ETC: 02:24 (0:00:00 remaining)
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb2-time:
|   date: 2026-08-13T06:24:37
|_  start_date: N/A
|_ clock-skew: 5s

Nmap done: 1 IP address (1 host up) scanned in 20.82 seconds
```

7. sudo nmap -p 22,23,80,443,161 -sV 192.168.56.1

```
(kali㉿kali)-[~]
$ sudo nmap -sC 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:24 -0400
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 93.92% done; ETC: 02:24 (0:00:00 remaining)
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3306/tcp   open  mysql

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb2-time:
|   date: 2026-08-13T06:24:37
|_  start_date: N/A
|_ clock-skew: 5s

Nmap done: 1 IP address (1 host up) scanned in 20.82 seconds
```

8. `sudo nmap -sU -p 161 192.168.56.1`

```
(kali㉿kali)-[~]  
$ sudo nmap -sU -p 161 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:25 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00091s latency).  
  
PORT      STATE      SERVICE  
161/udp    filtered   snmp  
  
Nmap done: 1 IP address (1 host up) scanned in 0.66 seconds
```

9. `sudo nmap -sU -p 161 --script snmp-info`

```
(kali㉿kali)-[~]  
$ sudo nmap -sU -p 161 --script snmp-info  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 02:26 -0400  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.06 seconds
```